# Real vs Fake Image and Video Detection

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# Introduction

- Deepfakes are AI-generated fake media
- Hard to detect manually
- Used in fraud and misinformation

# Problem Statement

- Rapid growth of fake content
- Manual detection unreliable
- Need automated AI system

# Objective

Detect real vs fake images/videos

High accuracy classification

Support video frame analysis

# Dataset

COCO 2017 → Real images

CIFAKE → Fake images

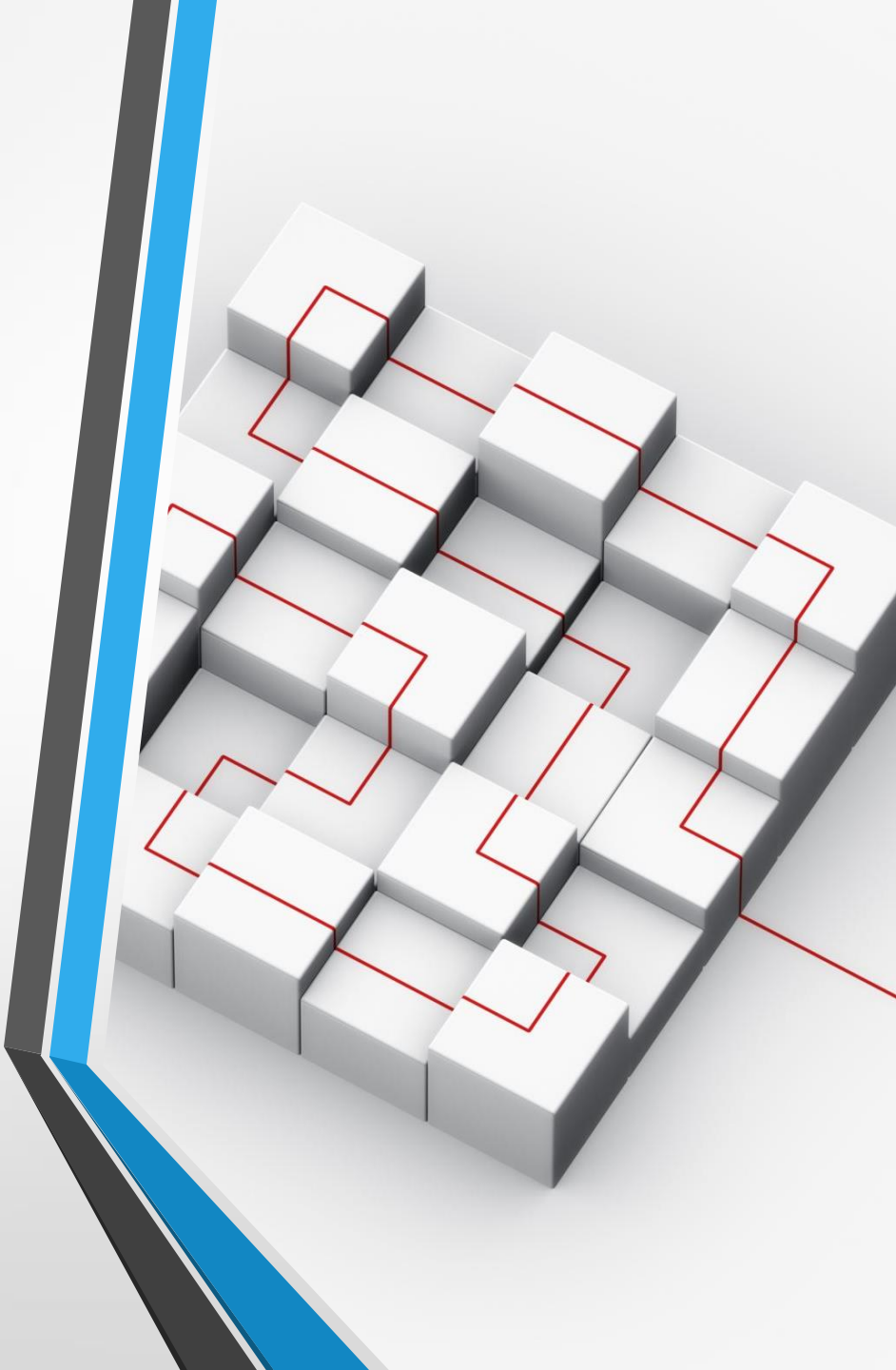
Binary classification (0 = Real, 1 = Fake)

# Methodology

**Input → Preprocessing → CNN → Prediction → Output**

# Model Architecture

- EfficientNet (Main model)
- ResNet (Supporting model)
- CNN-based feature extraction



# Working

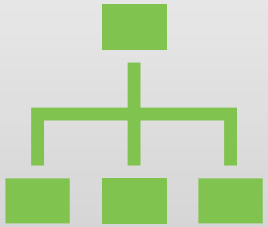
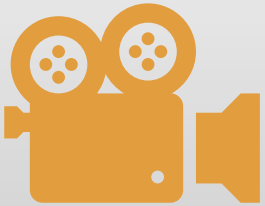


Image: Direct classification



Video: Frame extraction + majority voting

# System Flowchart

- Input Image/Video
- Preprocessing
- Feature Extraction (CNN)
- Prediction
- Final Output

# Results & Accuracy

EFFICIENTNET: TRAIN ~94%, VAL ~90%

RESNET: TRAIN ~92%, VAL ~87%

IMAGE ACCURACY: ~90%

VIDEO ACCURACY: ~88%

# Thankyou